Daniel Zhao

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Education

University of California, San Diego | San Diego, CA — SEP 2024 - DEC 2025

M.S. in Computer Science (AI focus) **GPA:** 3.90

Relevant Coursework: Deep Learning, Machine Learning, NLP, ML Systems, Artificial Intelligence, Recommender Systems, Computer Arch

University of California, Berkeley | Berkeley, CA — AUG 2020 - MAY 2024

B.A. in Computer Science & B.A. in Applied Math **GPA:** 3.90

Professional Experience

Member of Technical Staff | ***Axilon*** — DEC 2025 - PRESENT

- Ontologizing your factory; building agentic harnesses and AI-native knowledge graphs for the industrial world

Applied ML Engineer (Contract) | ***Outerproduct*** — FEB 2026 - APR 2026

- Distributed training and steering of LLMs (Mistral, Llama) using recursive feature machines to improve performance & interpretability

Machine Learning Engineer Intern | ***Qualcomm*** — JUN 2025 - SEP 2025

- Architect **multimodal RAG** system through Langchain, using ONNX-based CLIP embeddings, FAISS vector store, Qwen2.5VL
- Deploy system on Qualcomm-native CPU machine, benchmarking against CUDA host on Haystacks using LLM-as-a-judge
- **Cross-compile Llama.cpp** for Adreno GPU and quantize Qwen2.5 VL model to Q4_0, achieving over **10x speedup** from baseline CPU
- Extensively profile and fix CLIP projection encoding bottleneck in Llama.cpp C++ binaries, achieving 5x speedup from naive solution

Teaching Assistant | ***CSE 234: ML Sys, CSE 253: ML for Music, CSE 258: Recommenders @ UCSD*** — DEC 2024 - DEC 2025

- Constructed 3 brand-new assignments for students covering mat-mul optimization, backprop graph construction, speculative decoding
- Hosted 50+ office hours, proctor exams, hold discussion sections, maintain Ruby website under Prof. Hao Zhang & Prof. Julian McAuley

Research Intern | ***MUSAIC, Hao AI Lab @ UCSD*** — APR 2024 - DEC 2025

- Explore interpretability of music and guided real-time music generation through recursive feature machines and sparse autoencoders
- Configured new CLLM models to be compatible with **qLoRA fine-tuning** on Jacobi trajectories, decreasing memory usage over 3x

Machine Learning Engineer Intern | ***Accretional*** — JUN 2024 - SEP 2024

- Designed and implement **RAG vector** embedding model to ground LLM with respect to cloud infrastructure
- Fully constructed VSCode application in Svelte and TypeScript, creating tool-calling workflow in the **agentic AI** pipeline
- Created command-line tool in Golang for indexing and vectorization of files, **semantic & lexical search** of keywords

Software Engineer Intern | ***Hinge Health*** — MAY 2023 - AUG 2023

- Designed 5+ endpoints and event consumers in TypeScript and Ruby on Rails, restructuring and building new GraphQL queries
- Presented solutions for over 10 UI/UX features in React, linked repos with RabbitMQ event emitting, deployed in Kubernetes

Full Stack Intern | ***Cisco Meraki*** — MAY 2022 - AUG 2022

- Spearheaded a full stack project to create uplink ping stat graphs, helping **500,000 customers** visualize wireless trends over time
- Collected ping stats data from over 4 million wireless internet nodes, using Scala scripts to build a grabber and Jenkins to test

Research

Steering Autoregressive Music Generation with RFMs **(ICLR 2026)** — JAN 2026

- Interpretable classification of music using **recursive feature machines** (based on kernel ridge regression & AGOP) as classifiers
- Inject direction vectors into LLM's logits layer to steer LLM towards a musically interpretable concept (e.g. major vs. minor)

VideoScience-Bench ***(ECCV 2026 pending),*** an auto-eval physics reasoning benchmark for video generation — DEC 2025

CoT Reasoning with Sparse Autoencoders, clustering on SAE-based token rep for LLM interpretability — FEB 2025

Skills

Python, PyTorch, TensorFlow, Machine Learning, Artificial Intelligence, TypeScript, JavaScript, JupyterNotebook, Golang, SQL, C, C++, Kubernetes,React, Java, Unix, Git, Docker, AWS, Ruby on Rails, Scala, RabbitMQ, GraphQL, HTML, CSS, C#, Agentic Tech